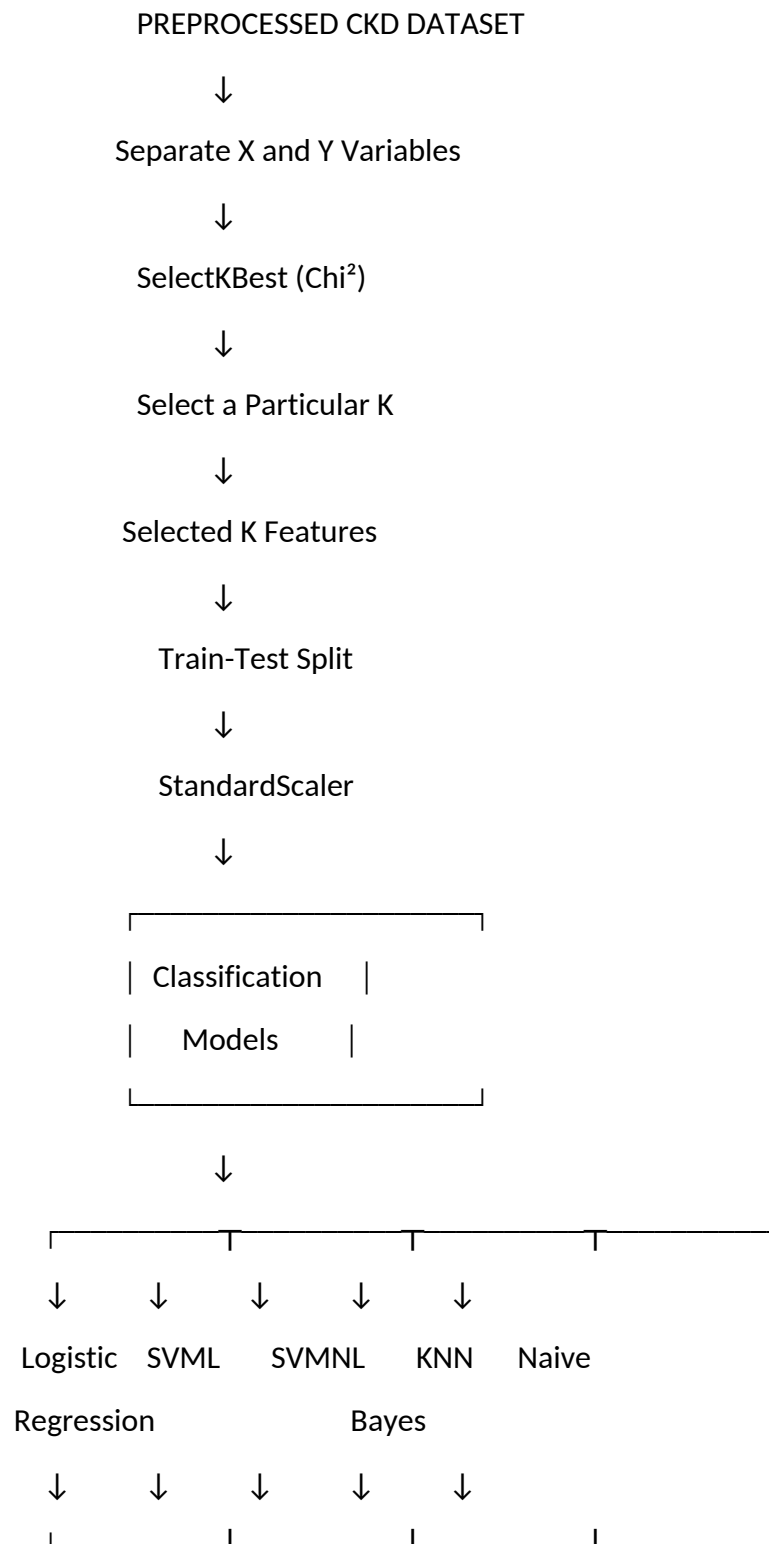


CHOOSING THE BEST MODEL BASED ON 'K' VALUE and MODEL's ACCURACY

Flow Chart of the Program:





Decision Tree



Random Forest



Prediction on

Test Data



Confusion Matrix



Calculate Accuracy



Accuracy of Each Model	
Logistic Regression	
SVML	
SVMNL	
KNN	
Naive Bayes	
Decision Tree	
Random Forest	



Create Accuracy DataFrame



MODEL	ACCURACY
Logistic Regression	95%

	SVML	94%	
	SVMNL	96%	
	KNN	93%	
	Naive Bayes	91%	
	Decision Tree	92%	
	Random Forest	97%	



Compare Model Accuracy



Select Best Performing

Model

Comparison Table of each Model's Accuracy:

K Value	Model Name	Accuracy	Highest
3	DecisionTree	84%	DT-3
3	RandomForest	83%	
4	DecisionTree	89%	DT-4 and RF-4
4	RandomForest	89%	
5	DecisionTree	96%	DT-5
5	RandomForest	95%	
6	DecisionTree	97%	DT-6 and RF-6
6	RandomForest	97%	
7	DecisionTree	98%	DT-7
7	RandomForest	97%	

Result:

From the above table, we can decide that 'DecisionTree' model's Accuracy is highest and it is consistently higher than other model's Accuracy value and the corresponding 'k' value is 7.

Best Model - DecisionTree; K value - 7